

MIKHAIL BIN YASSIN

Data Engineer · ETL & Big Data · Workflow Orchestration

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PROFESSIONAL SUMMARY

Data Engineer with a Bachelor of Computer Science in Data Engineering and nine months of hands-on experience building and deploying PySpark and Python ETL pipelines for telecom billing, activation, and reconciliation reporting. Experienced in processing approximately 2 billion CDR rows per day, investigating billing discrepancies, and automating operational workflows on an on-premise big-data platform. Built a DAG-based ETL scheduling system as a final-year project and developed full-stack web applications with real users. Comfortable working across SQL, Python, PySpark, Linux, Bash, ClickHouse, PostgreSQL, Git, and cloud data workflows.

CORE COMPETENCIES

Data engineering: ETL development · big-data processing · workflow orchestration · database design · data analysis & reporting · system design

Tools & technologies: Python · PySpark · SQL · PostgreSQL · ClickHouse · Bash/Shell · Linux · Git · OpenTelemetry · AWS

Platforms & interests: real-time data processing · cloud data systems · observability dashboards · agentic engineering · applied AI

PROFESSIONAL EXPERIENCE

U Mobile — Data Science & Campaign Value Management

Johor Bahru, Malaysia · Nov 2025 – Aug 2026

Data Engineering Intern · 9 months

- Designed, developed, and deployed **PySpark and Python ETL pipelines** for billing, activation, and reconciliation reporting on an on-premise big-data platform, covering requirements gathering, UAT, and production deployment.
- Processed approximately **2 billion CDR rows per day** with PySpark to prepare curated datasets for a cloud DaaS proof of concept with an external vendor.
- Investigated inter-operator billing discrepancies and settlement reports, identifying root causes and supporting data-driven resolution.
- Improved operational automation through secure cloud file transfers, variance-alert emails, and file-permission management.
- Contributed to a centralised platform-monitoring dashboard proof of concept using **ClickStack**.
- Supported legacy-platform migration, server operations, and internal knowledge-sharing sessions.

SELECTED PROJECTS

Design & Implementation of a DAG-based ETL Scheduling System

Final-Year Project · 2025–2026

Research and engineering project — Universiti Teknologi Malaysia

- Designed a custom pipeline-orchestration system that models ETL dependencies as a directed acyclic graph and dispatches tasks when their dependencies complete.
- Implemented graph validation, topological ordering, and completion-based dispatch, with automated graph and schema tests.
- Produced a public research summary covering the scheduling approach and system design; full end-to-end runtime validation remains pending.

Kindergarden Management System — Intellect Playschool

Work-Based Learning · 2024

University Work-Based Learning project · Team Lead

- Led a four-person team in developing a web application for Intellect Playschool, a kindergarten in Pulai Mutiara, Johor Bahru, under university supervision.
- Contributed to the application design and delivery for a real education centre; the team was selected as the **Best Group** for the semester.
- Public repository: github.com/mikhaily/Kindergarden-Management-System

RehabConnect — Autism Education Centre Management System

Work-Based Learning · 2025

University Work-Based Learning project · Team Lead

- Led a four-person team in building a web application for a local autism education centre in Johor Bahru.
- Managed technical delivery and knowledge sharing; the project was supervised by Mr. Syukry Radzi, CEO of Cloudswired Technologies.
- The team was selected as the **Best Group** for the semester. Public repository: github.com/mikhaily/RehabConnect

Omarchy → BB Theme Sync

Open source · 2026

Open-source systems integration

- Designed a Bash integration that maps Omarchy `colors.toml` values to BB custom CSS themes and activates them through the `theme-set.d` hook.
- Implemented atomic stylesheet writes, scoped symlink installation, safe uninstall behaviour, light/dark

palette handling, and ApplImage/packaged BB CLI discovery.

- Added an isolated verification suite, GitHub Actions CI, contribution guidelines, issue templates, security policy, and a tagged **v0.1.0** release.

EDUCATION

Universiti Teknologi Malaysia, Johor Bahru

Oct 2021 – Oct 2026

Bachelor of Computer Science (Data Engineering) · CGPA 3.85

Universiti Institusi Teknologi MARA, Dengkil

Jul 2020 – May 2021

Foundation in Engineering · CGPA 4.00

CERTIFICATIONS

- AWS Academy Data Engineering
- AWS Academy Cloud Architecting
- Google AI Professional
- Microsoft Certified: Azure Data Fundamentals

LEADERSHIP & ACTIVITIES

- **Faculty of Computing Research & Innovation Day (FCRI 2025) — Showcase Planning Team Lead:** Led five people handling communications with Finexus Group Sdn. Bhd. for a potential product showcase booth.
- **Sukan Fakulti Komputeraan 2023 (SUSKOM'23) — Multimedia Team Lead:** Led five people in multimedia content and event coverage.
- **Kolej Tun Dr Ismail Earth Day 2023 — Multimedia Team Lead:** Supervised a five-person team delivering promotional multimedia materials.

LANGUAGES

Malay — Native · English — Proficient